Supplementary Materials

Perceived Stress Tracks the Tinnitus Symptom More Than the Audiometric Threshold: A Survey-Weighted KNHANES Study (STARS)

These Supplementary Materials collect (i) the *prospective* program components of STARS (design only; no results in the main empirical article) and (ii) the methods-detail tables. They are referenced from the main text as "Supplementary Methods S1/S2" and "Supplementary Table S1–S3."

Supplementary Methods

S1. Prospective external validation (companion program)

Cross-national transportability of a *fitted* KNHANES model to NHANES (Aim 2) is a prospective component and carries no results in the main empirical article; the NHANES analyses reported in the main text are directional support under a distinct distress proxy, not a validation of a transported model. To keep the plan falsifiable and auditable, the transportability protocol is prespecified and recorded in the repository: weighted versus unweighted performance; calibration-in-the-large and slope before and after intercept-only then intercept+slope recalibration of the frozen model; common-support restriction to the overlapping covariate region; case-mix standardization to separate genuine transportability from prevalence/age differences; and a reduced common model versus country-specific extended models. Discrimination and calibration are prespecified to be reported within subgroups (sex, age band, employment, noise exposure, unilateral/asymmetric hearing, socioeconomic status, and race/ethnicity where collected), with subgroup calibration and error gaps as first-class results. These analyses are developed in the companion program of work.

S2. Prospective clinical extension (hospital-based cohort)

A prospective otolaryngology/audiology cohort is specified to address what public cross-sectional data cannot: treatment timing, sudden sensorineural hearing loss (SSNHL) recovery trajectories, tinnitus burden over time, and hearing-aid rehabilitation. This cohort contributes deidentified audiograms and clinical narratives for clinical external validation and provides the substrate for the AI extraction and safety evaluations developed in the companion SAFE-EAR paper. Raw clinical data are not redistributed; only derived, deidentified, or synthetic artifacts are shared through the open repository, subject to institutional review board (IRB) approval and data-governance review.

Supplementary Tables

Table S1: Selected public datasets for STARS

Dataset	Role	Strengths	Main limitations
KNHANES	Primary development dataset	Korean population relevance; audiometry in selected cycles; tinnitus and health variables; survey weights	Cross-sectional; stress variables may be perceived stress rather than validated occupational stress; raw data access procedures required
NHANES	External validation dataset	Public CDC files; audiometry and noise exposure in selected cycles; rich health covariates; geographic validation	U.S. population; cycle-specific hearing and tinnitus variables; complex survey design
OHHR / Zenodo audiology data	Secondary method testing	Audiology-specific measurements; useful for model benchmarking	May lack stress, occupation, or tinnitus variables
OpenNeuro tinnitus datasets	Optional neurophysiology substudy	Public neuroimaging or fNIRS data for tinnitus biomarkers	Not designed for occupational stress or longitudinal treatment outcomes

Table S2: Target harmonized variables, roles, and availability. Exposure and outcome roles follow the naming principle and the main-text DAG figure: perceived stress is the exposure; work-related factors and the minimal confounder set are covariates; depressed mood and sleep are treated as mediators in the primary analysis. Availability is cycle-dependent; see Table S3 for source items and comparability.

Harmonized variable	Role	KNHANES	NHANES
Tinnitus presence	Outcome	Sel. cycles	Sel. cycles
Bothersome tinnitus	Outcome	Sel. cycles	Sel. cycles
Better-ear $PTA_{0.5,1,2,4}$	Outcome	Sel. cycles	Sel. cycles
Worse-ear PTA & per-ear thresholds	Outcome	Sel. cycles	Sel. cycles
Asymmetric hearing loss	Outcome	Sel. cycles	Sel. cycles
Self-reported hearing difficulty	Outcome (2^{nd})	Yes	Yes
Hearing-aid use	Outcome (2^{nd})	Sel. cycles	Sel. cycles
Perceived stress	Exposure	Yes	Yes
Work-related factors (occupation, employment, hours)	Covariate / modifier	Yes	Yes
Occupational + recreational noise (duration/intensity)	Confounder	Sel. cycles	Yes
Depressed mood	Mediator	Yes	Yes
Sleep duration	Mediator	Sel. cycles	Sel. cycles
Smoking, alcohol, cardiometabolic disease	Confounder	Yes	Yes
Age, sex, education	Confounder	Yes	Yes
Socioeconomic status; race/ethnicity	Fairness axis	Partial	Yes
Survey weight, stratum, PSU	Design	Yes	Yes

Table S3: Cross-national harmonization and measurement-comparability plan (excerpt). Exact cycle-specific variable codes and response categories are pinned in the repository data dictionary. Comparability governs use: *low*-comparability variables are excluded from the common transportable model and entered only in country-specific extended models. "Sel. cycles" = present in selected cycles only.

Harmonized variable	KNHANES (source/cycle)	NHANES (source/cycle)	Original item (abridged)	Transformation rule	Compar.
Tinnitus presence	Otologic Q, 2010–2012	Audiometry Q (AUQ), sel. cycles	"Ringing/noise in ears past 12 mo?"	Any vs. none (binary)	High
Bothersome tinnitus	Otologic Q, 2010–2012	AUQ bother item, sel. cycles	Annoyance / sleep / concentration	Bothersome vs. not	Medium
Pure-tone thresholds	Audiometry exam, 2010–2012	Audiometry exam, sel. cycles	Air-conduction dB HL per freq.	Better/worse-ear PTA; per-ear	High
Perceived stress	Health Q (stress item)	DPQ / stress-related item	"How much stress in daily life?"	Collapse to high vs. low	**Low**
Depressed mood	PHQ-9 / mood item	PHQ-9 (DPQ)	Standardized depression items	Screen-positive vs. negative	Medium
Occupation / employment	Socioecon. Q	Occupation (OCQ)	Job category, employed y/n	Common category crosswalk	Medium
Occupational noise	Sel. cycles	Noise exposure (AUQ)	Loud-noise exposure, duration	Duration/intensity + binary	Medium
Sleep duration	Health Q	Sleep (SLQ)	Average hours of sleep	Continuous hours	Medium
Smoking / alcohol	Health Q	SMQ / ALQ	Current status, quantity	Standard categories	High
Age, sex, education	Demographics	Demographics (DEMO)	—	Direct / crosswalk	High
Survey weight, stratum, PSU	Design vars	Design vars	—	Kept for design-consistent SEs	High